

LAPORAN PRAKTIKUM

Pemrograman Website - Bab 6: Framework UI React JS Part 2

CMS Books Management (Bacapedia+)

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1. Tujuan

- Membangun modul CMS (Content Management System) untuk manajemen data buku.
- Menerapkan Sidebar Navigation, Props, State, dan Hooks secara terintegrasi.
- Membuat komponen UI global yang reusable (Button, Alert, Forms, Cards, Modal, Datatable).
- Mengimplementasikan form input buku dan modal Add New Book.

2. Struktur Project

src/app/cms/ (layout.jsx, page.jsx, books/page.jsx) -> routing halaman CMS

src/components/cms/ (_books, components/sidebar.jsx, components/card_calculates.jsx)

src/components/ui/ (buttons, alert, forms, cards, modals, datatables)

src/const/ (bookList.jsx, menu_cms.jsx)

src/hooks/useScrolled.jsx

3. Penjelasan Komponen

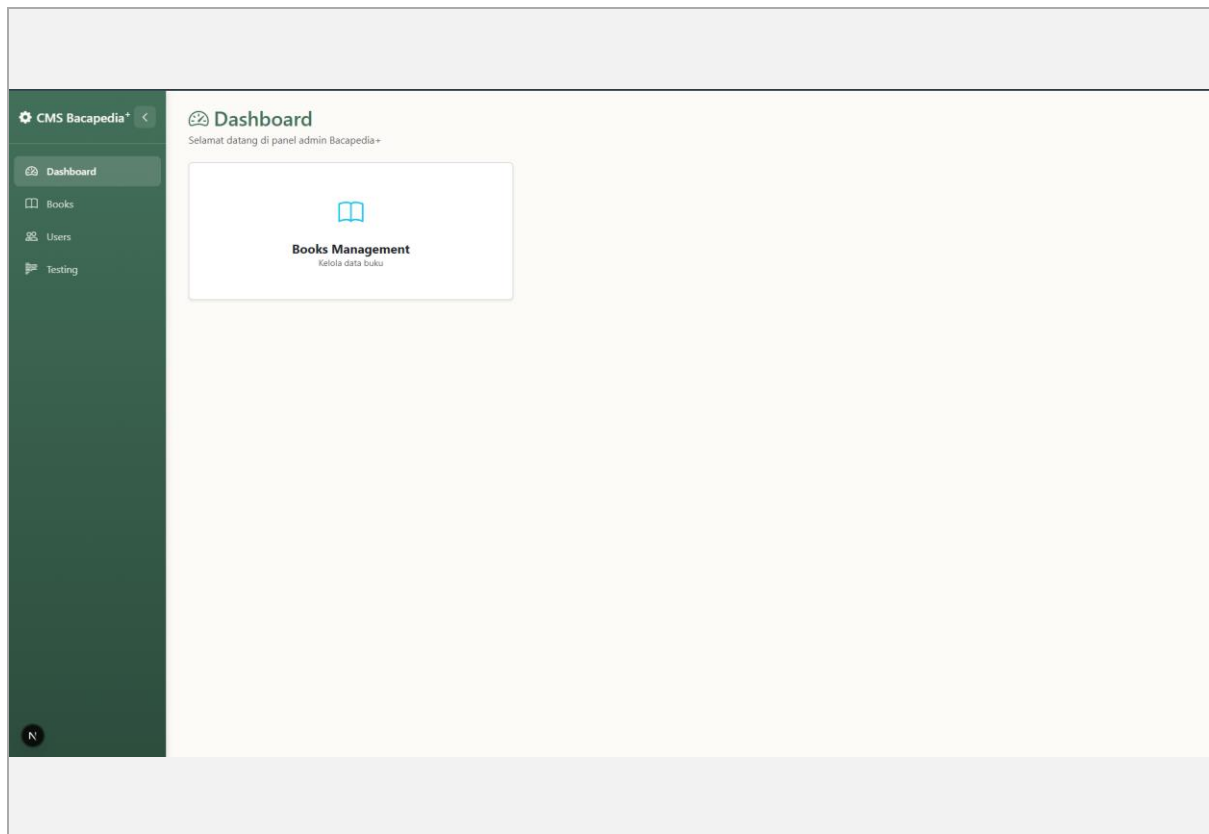
- Sidebar: navigasi collapse/expand dengan deteksi active route (usePathname) dan hover-to-expand.
- CardCalculates: menampilkan ringkasan statistik (total buku, gratis, subscribe) di dashboard.
- Tabledata: tabel buku dengan fitur search, sorting, dan pagination memakai useMemo.
- Form: form input buku (Book Title, Author, Type Book, Sinopsis, Story, Upload Image) memakai komponen ui/forms.jsx.
- Modals: popup global berbasis Singleton State, dipanggil dari mana saja lewat openModal().
- Header: judul halaman + tombol Add New Book yang memicu openModal() menampilkan Form.

4. Perbaikan Bug yang Ditemukan

- tabledata.jsx: handleEdit dan handleDelete dipanggil tapi belum didefinisikan -> ditambahkan fungsi agar tombol Edit/Delete tidak error.
- modals.jsx: tombol "No" pada modal generik memanggil handleEdit(book) yang tidak ada di scope -> diarahkan ke data.onClose.
- app/cms/page.jsx: file kosong tanpa default export -> diisi halaman dashboard sederhana.

5. Screenshot Tampilan Aplikasi

Halaman Dashboard CMS (/cms):



Halaman Books Management (/cms/books) - tabel, search, sorting, pagination:

CMS Bacapedia*

Dashboard

Books

Users

Testing

Books Management

Manage your book collection

Add New Book

Total Books6

Free Book5

Subscribe1

Subscribe6

Book Lists

Search here...

NO	TITLE	AUTHOR	LANGUAGE	RATE/VIEW	SUBSCRIBE	ACTIONS
1	Harry Potter and the Sorcerer's Stone	J.K. Rowling	English	★ 4.5 1000	No	 
2	Hunger Games	Suzanne Collins	English	★ 4.9 2000	No	 
3	Maze Runner	James Dashner	English	★ 4.7 500	No	 
4	Divergent	Veronica Roth	English	★ 3.7 300	No	 
5	1984	George Orwell	English	★ 4.8 1500	Yes	 
6	The Hobbit	J.R.R. Tolkien	English	★ 4.6 800	No	 

CMS Bacapedia*

Dashboard

Books

Users

Testing

Books Management

Manage your book collection

Add New Book

Total Books6

Free Book5

Subscribe1

Subscribe6

Book Lists

Harry

NO	TITLE	AUTHOR	LANGUAGE	RATE/VIEW	SUBSCRIBE	ACTIONS
1	Harry Potter and the Sorcerer's Stone	J.K. Rowling	English	★ 4.5 1000	No	 

Edit Book

Fill in the details for the book.

Demo form only - not connected to API

Book Title *

Author Name *

Type Book *

Is Free

Sinopsis *

Cover Image *

Choose File

No file chosen

Story *

Cancel

Submit Book

Modal Add New Book saat tombol diklik:

Add New Book

Fill in the details for the book.

Demo form only - not connected to API

Book Title *

Author Name *

Type Book *

☐ Is Free

Sinopsis *

Cover Image *

Choose FileNo file chosen

Story *

Cancel

Submit Book

6. Screenshot Code yang Dimodifikasi

src/app/cms/ (layout.jsx, page.jsx,) -> routing halaman CMS

```
my-first-app-react > src > app > cms > layout.jsx > ...
1  import Sidebar from "@components/cms/components/sidebar";
2  import Modals from "@components/ui/modals";
3  import "./cms.css";
4
5  export default function CMSLayout({ children }) {
6    return (
7      <>
8        <div className="cms-container">
9          <Sidebar />
10         <main className="main-content p-4">{children}</main>
11        </div>
12        <Modals />
13      </>
14    );
15  }
16
```

```
CodeSnap booklist.jsx page.jsx X
my-first-app-react > src > app > cms > page.jsx > ...
1 import Link from "next/link";
2
3 export default function CMSDashboard() {
4   return (
5     <div className="container-fluid">
6       <h2 className="mb-1 text-green">
7         <i className="bi bi-speedometer2 me-2"></i>
8         Dashboard
9       </h2>
10      <p className="text-muted mb-4">Selamat datang di panel admin Bacapedia+</p>
11      <div className="row g-4">
12        <div className="col-md-4">
13          <Link href="/cms/books" className="text-decoration-none">
14            <div className="card h-100 shadow-sm">
15              <div className="card-body text-center py-5">
16                <i className="bi bi-book fs-1 text-info"></i>
17                <h5 className="fw-bold mt-3 mb-0">Books Management</h5>
18                <p className="text-muted small mb-0">Kelola data buku</p>
19              </div>
20            </div>
21          </Link>
22        </div>
23      </div>
24    </div>
25  );
26 }
```

src/components/cms/(_books, components/sidebar.jsx, components/card_calculates.jsx)

EXPLORE

PW-MERN-STACK-242310026

my-first-app-react

src

components

cms

_books

components

formjsx

headerjsx

tabledatajsx

indexjsx

pagejsx

components

indexjsx

landing-pages

ui

indexjsx

const

booklistjsx

menu_cmsjsx

navMenujsx

hooks

.gitignore

eslint.config.mjs

TS next-env.d.ts

TS next.config.ts

package-lock.json

package.json

README.md

tsconfig.json

CodeSnap

booklistjsx

formjsx

my-first-app-react > src > components > cms > _books > components > formjsx > ...

```
1 "use client";
2
3 import React, { useState } from 'react'
4 import { TextAreaInput, TextInput, InputImage, InputCheckbox } from '@components/ui/forms'
5 import { Button } from "@components/ui/button";
6 import { Alert } from '@components/ui/alert';
7
8 export default function Form({ book_id, ReloadBook }) {
9   const obj_book = {
10     title: '',
11     author: '',
12     synopsis: '',
13     story: '',
14     is_free: false,
15     image: null
16   }
17   const [formData, setFormData] = useState(obj_book)
18   const [imagePreview, setImagePreview] = useState(null)
19   const [error, setError] = useState("Demo form only - not connected to API");
20
21   const handleInputChange = (e) => {
22     const { name, value, type, checked } = e.target
23     setFormData(prev => ({
24       ...prev,
25       [name]: type === 'checkbox' ? checked : value
26     })))
27   }
28
29   const handleImageChange = (e) => {
30     const file = e.target.files[0]
31     if (file) {
32       const validTypes = ['image/jpeg', 'image/jpg', 'image/png']
33       if (!validTypes.includes(file.type)) {
34         setError('Please select a valid image file (JPEG, or PNG)')
35         return
36       }
37
38       if (file.size > 5 * 1024 * 1024) {
39         setError('Image size must be less than 5MB')
40         return
41       }
42
43       setFormData(prev => ({ ...prev, image: file })))
44       setError("")
45
46       const reader = new FileReader()
47       reader.onloadend = () => {
48         setImagePreview(reader.result)
49       }
50       reader.readAsDataURL(file)
51     }
52   }
53
54   const handleSubmit = async (e) => {
55     e.preventDefault()
56     setError("Demo form only - no save action is connected")
57   }
```

EXPLORER

my-first-app-react > src > components > cms > _books > components > tabledatajsx > ...

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🔍 📄 🔄 🗑

▼ backend

▼ my-first-app-react

▼ next

▼ node_modules

▼ public

▼ src

▼ app

▼ components

▼ cms

▼ _books

▼ components

form.jsx

header.jsx

tabledata.jsx

index.jsx

page.jsx

components

index.jsx

landing-pages

ui

index.jsx

const

booklist.jsx

menu_cms.jsx

navMenu.jsx

hooks

.gitignore

JS eslint.config.mjs

TS next-env.d.ts

TS next.config.ts

(!) package-lock.json

(!) package.json

(!) README.md

tsconfig.json

OUTLINE

TIMELINE

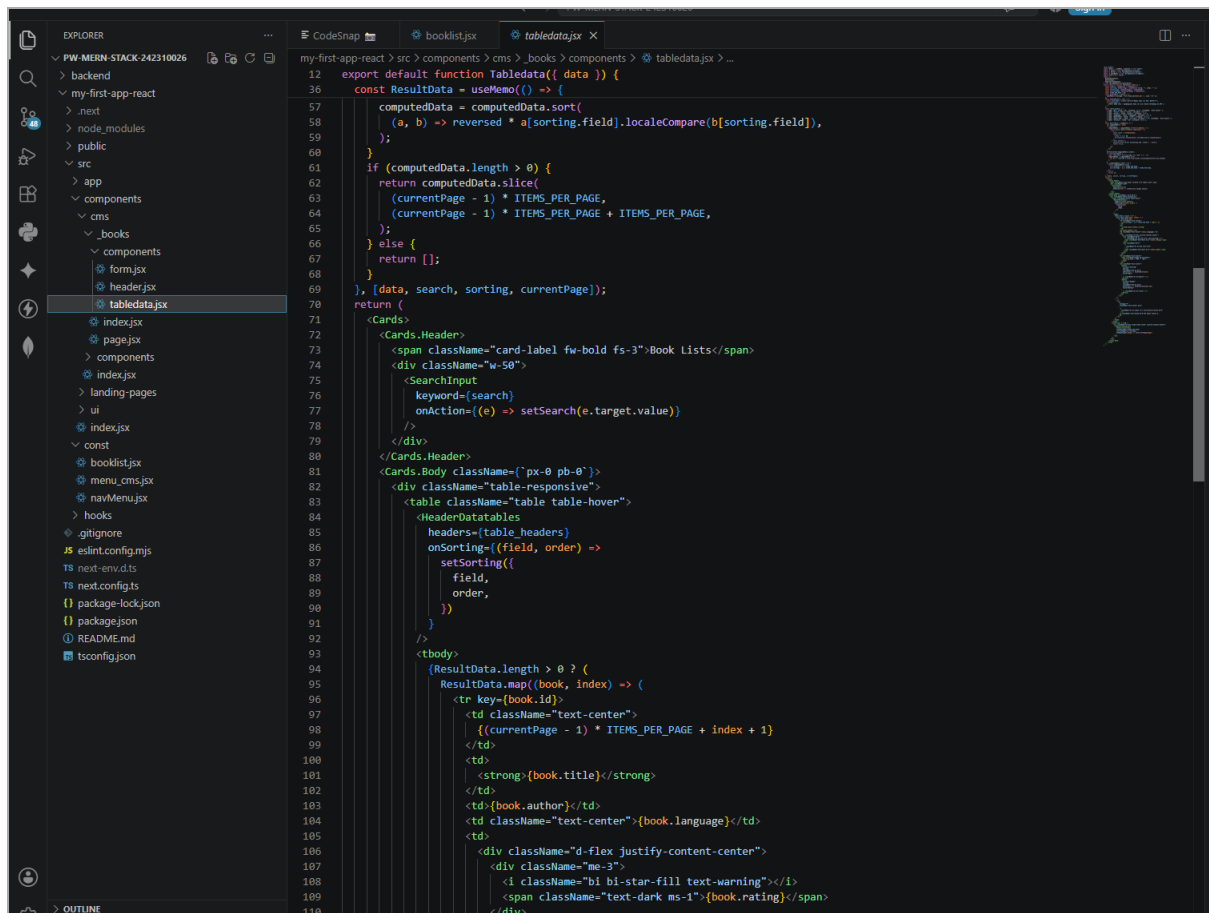
CodeSnap

booklist.jsx

tabledata.jsx

my-first-app-react > src > components > cms > _books > components > tabledatajsx > ...

```
1 "use client";
2 import React, { useMemo, useState } from "react";
3 import { Cards } from "@components/ui/cards";
4 import { Button } from "@components/ui/button";
5 import { openModal } from "@components/ui/modals";
6 import Form from "../form";
7 import {
8   HeaderDatatables,
9   SearchInput,
10  PaginationComponent,
11 } from "@components/ui/datatables";
12 export default function Tabledata({ data }) {
13   const [search, setSearch] = useState("");
14   const [sorting, setSorting] = useState({ field: "", order: "" });
15   const [totalItems, setTotalItems] = useState(0);
16   const [currentPage, setCurrentPage] = useState(1);
17   const ITEMS_PER_PAGE = 10;
18   const handleEdit = (book) => {
19     openModal({ message: <Form book_id={book.id} />, size: "xl" });
20   };
21   const handleDelete = (id) => {
22     const confirmed = window.confirm("Hapus buku ini dari daftar?");
23     if (confirmed) {
24       alert("Demo only - penghapusan buku id ${id} belum terhubung ke API");
25     }
26   };
27   const table_headers = [
28     { name: "No", field: "id", sortable: false, className: "text-center" },
29     { name: "Title", field: "title", sortable: true },
30     { name: "Author", field: "author", sortable: true },
31     { name: "Language", field: "language", sortable: true },
32     { name: "Rate/View", field: "rate", sortable: false },
33     { name: "Subscribe", field: "is_free", sortable: true, className: "text-center" },
34     { name: "Actions", field: "id", sortable: false },
35   ];
36   const ResultData = useMemo(() => {
37     let computedData = data;
38     if (search) {
39       computedData = computedData.filter((listData) => {
40         return Object.keys(listData).some((key) => {
41           try {
42             const value = listData[key];
43             return (
44               value != null &&
45               String(value).toLowerCase().includes(search.toLowerCase())
46             );
47           } catch (error) {
48             console.error("Error processing key `${key}`:", error);
49             return false;
50           }
51         });
52       });
53     }
54     setTotalItems(computedData.length);
55     if (sorting.field) {
56       const reversed = sorting.order === "asc" ? 1 : -1;
57       computedData = computedData.sort(
```



EXPLORER

my-first-app-react

src

components

cms

books

components

tabledata.jsx

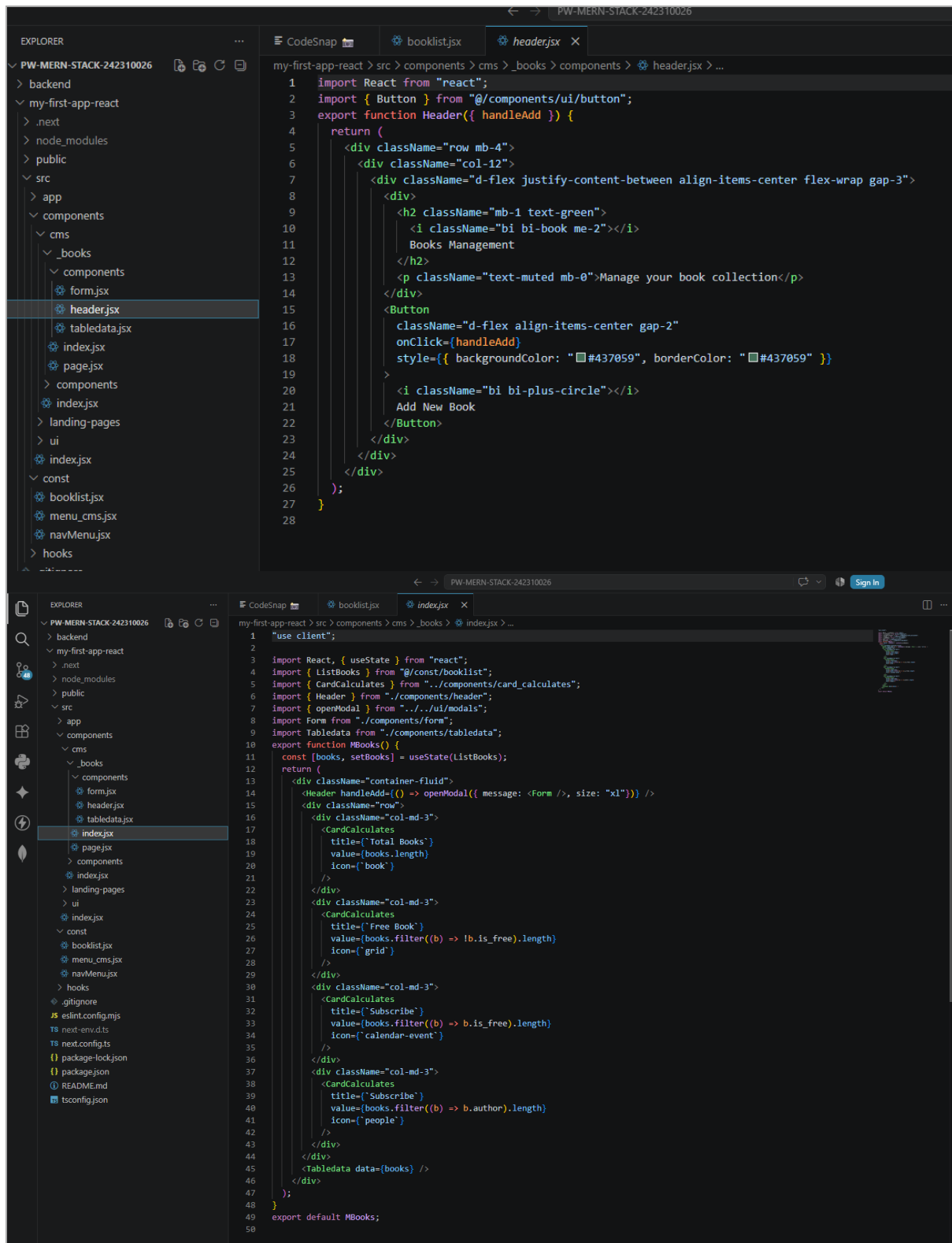
CodeSnap

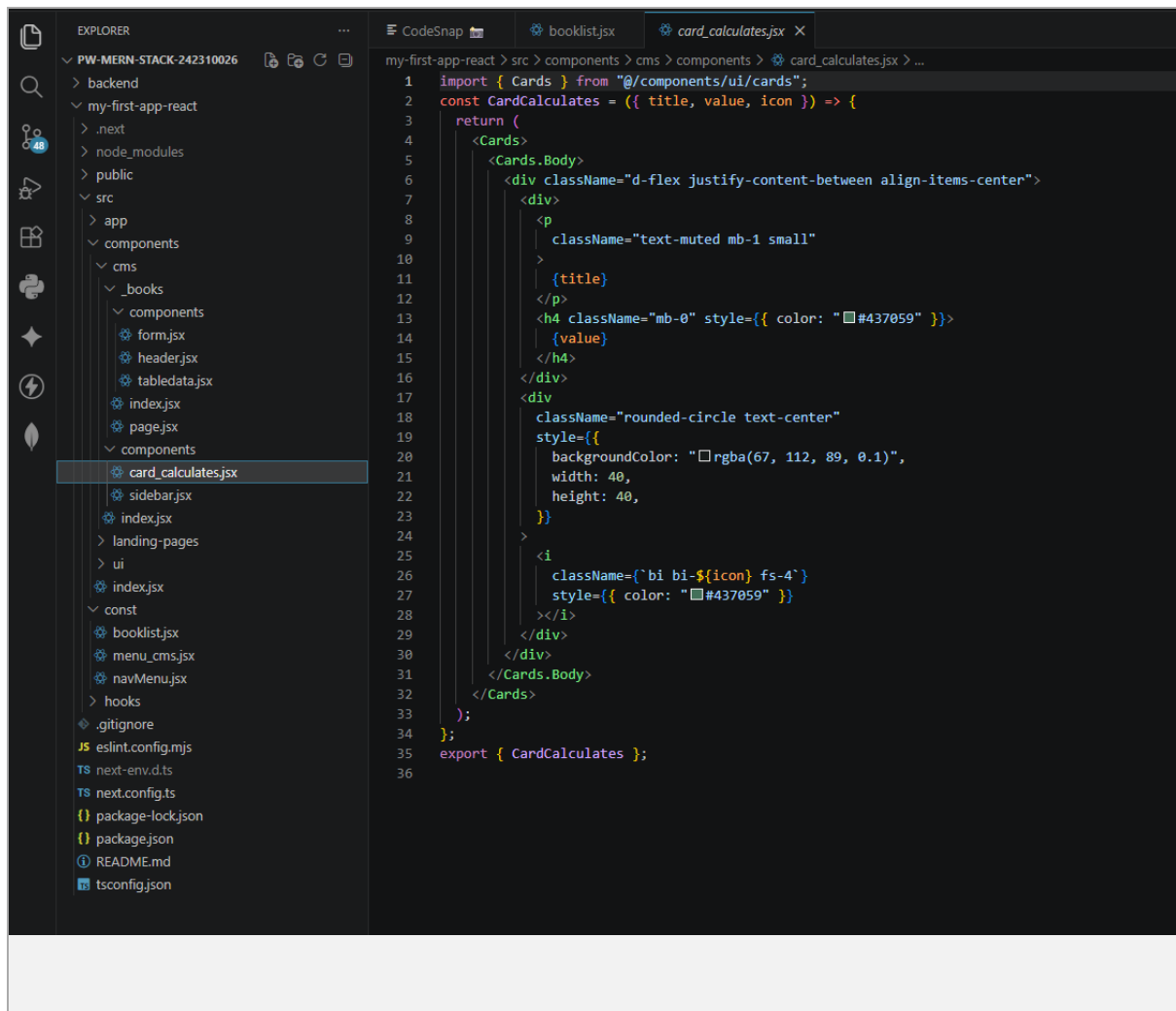
booklist.jsx

tabledata.jsx

```
12 export default function Tabledata({ data }) {
13   ResultData.map((book, index) => (
14     </td>
15     <td className="text-center">
16       <Button
17         variant="warning"
18         outline
19         className="btn-sm me-2"
20         onClick={() => handleEdit(book)}
21         title="Edit"
22       />
23       <i className="bi bi-pencil"></i>
24     </Button>
25     <Button
26       variant="danger"
27       outline
28       className="btn-sm me-2"
29       onClick={() => handleDelete(book.id)}
30       title="Delete"
31     >
32       <i className="bi bi-trash"></i>
33     </Button>
34   </td>
35 </tr>
36 ))
37 ) : (
38   <tr>
39     <td
40       colSpan="7"
41       className="text-center py-5"
42     >
43       <i
44         className="bi bi-inbox fs-1 text-muted d-block mb-3"
45       ></i>
46       <p className="text-muted mb-0">No books found</p>
47     </td>
48   </tr>
49 </tbody>
50 </table>
51 (totalItems > 0 && (
52   <div className="d-flex align-items-center justify-content-center">
53     <PaginationComponent
54       total={totalItems}
55       itemsPerPage={ITEMS_PER_PAGE}
56       currentPage={currentPage}
57       onPageChange={(page) => setCurrentPage(page)}
58     />
59   </div>
60 </div>
61 </Cards.Body>
62 </Cards>
63 );
64 }
```

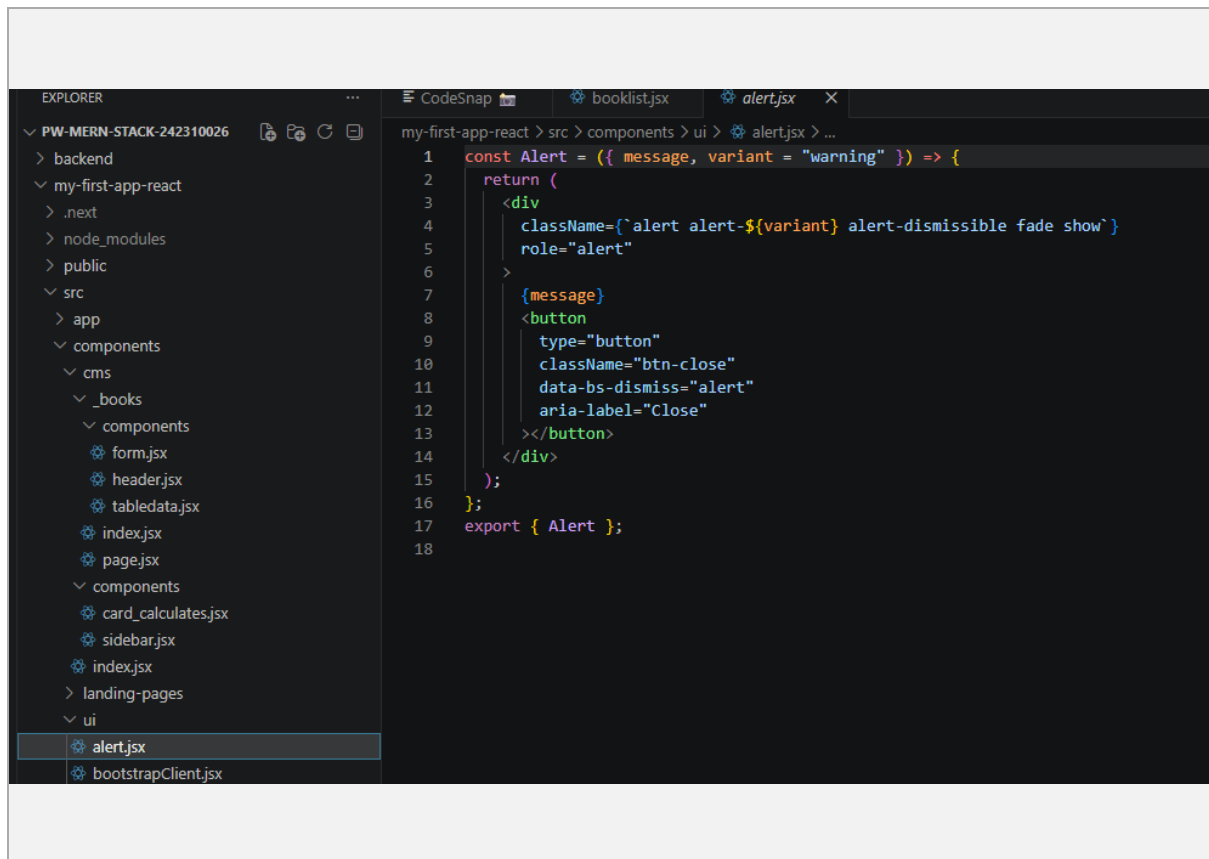
OUTLINE

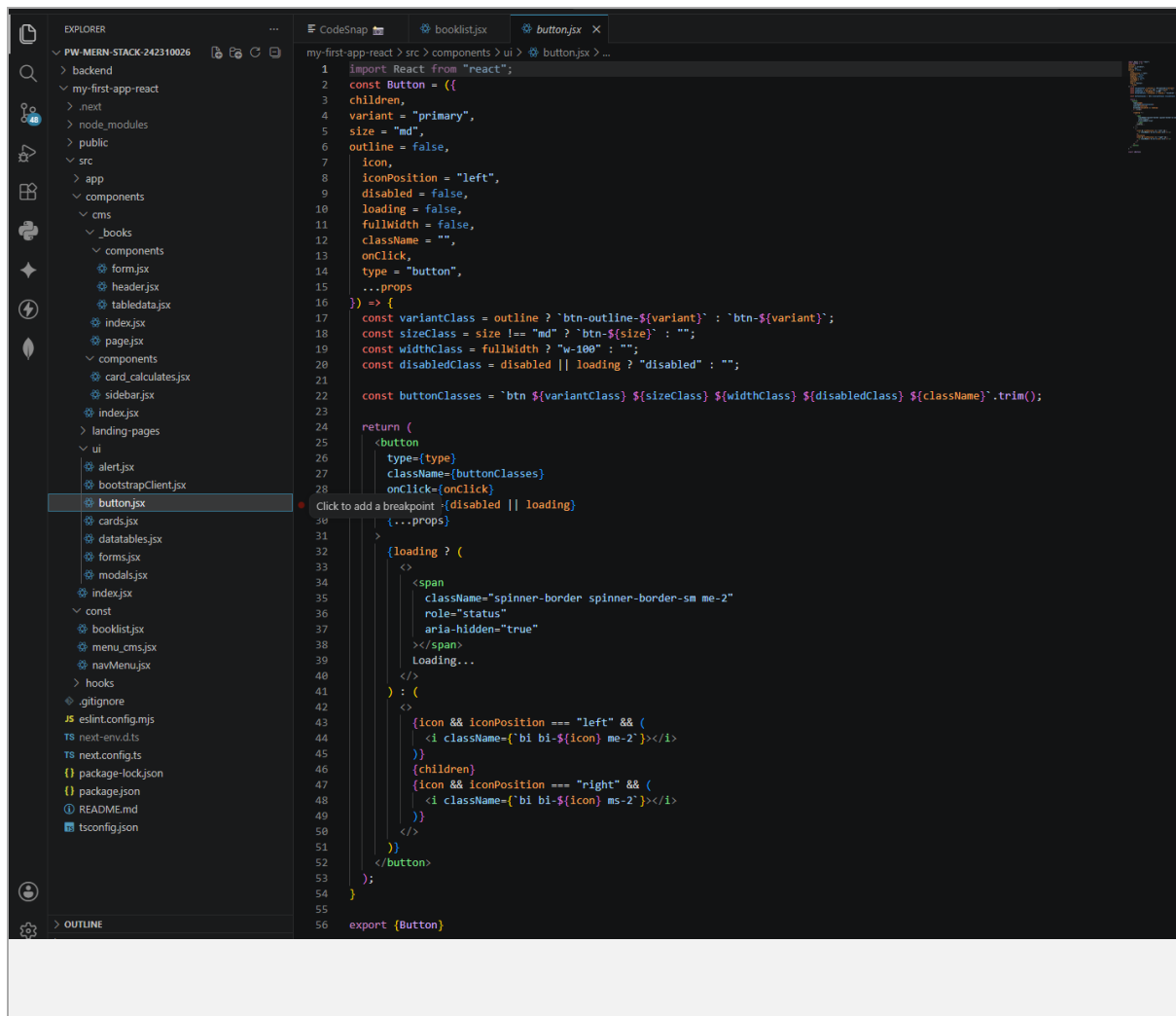


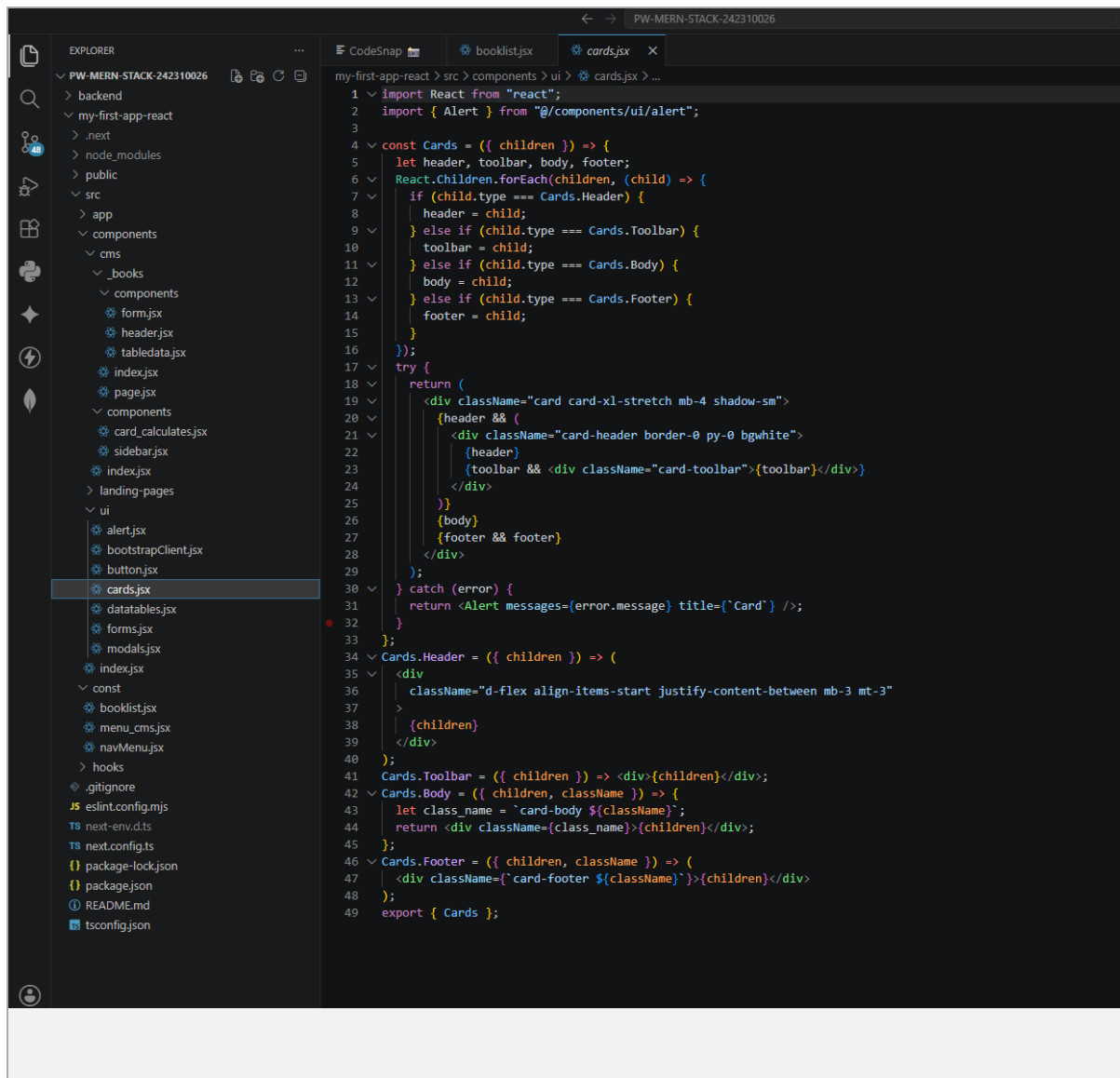


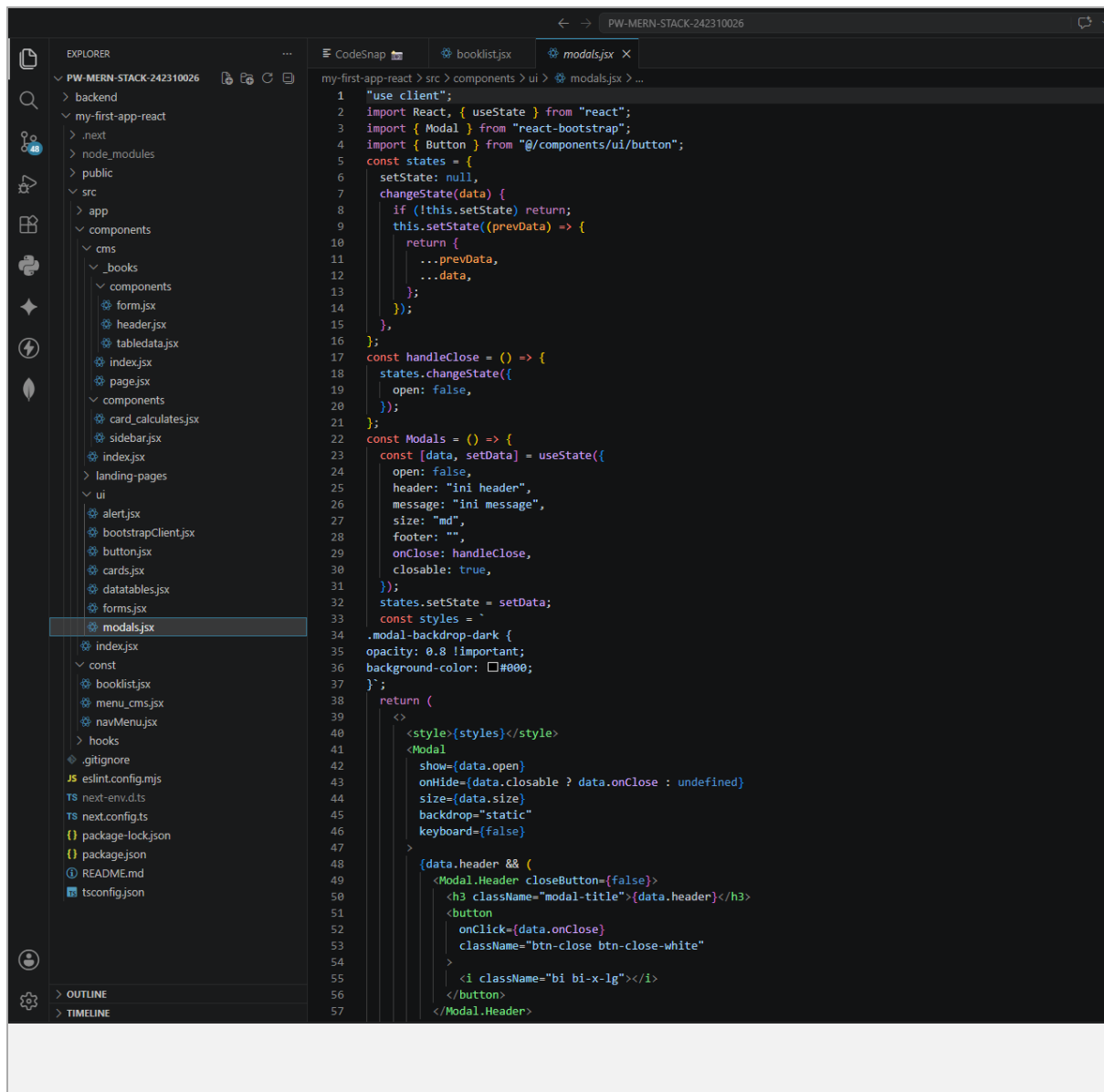
```
1  "use client"
2
3  import Link from 'next/link';
4  import { usePathname } from 'next/navigation';
5  import { useState } from 'react';
6  import { Menu_CMS } from '@/const/menu_cms';
7
8  export default function Sidebar() {
9    const pathname = usePathname();
10   const [isCollapsed, setIsCollapsed] = useState(false);
11   const [isHovered, setIsHovered] = useState(false);
12   const toggleSidebar = () => {
13     setIsCollapsed(!isCollapsed);
14   };
15   const handleMouseEnter = () => {
16     if (isCollapsed) {
17       setIsHovered(true);
18     }
19   };
20   const handleMouseLeave = () => {
21     setIsHovered(false);
22   };
23   const isExpanded = !isCollapsed || isHovered;
24   return (
25     <aside
26       className={`sidebar ${isExpanded ? "collapsed" : ""}`}
27       onMouseEnter={handleMouseEnter}
28       onMouseLeave={handleMouseLeave}
29     >
30       <div className="sidebar-header">
31         {isExpanded && (
32           <h4>
33             <i className="bi bi-gear-fill me-2"></i>
34             CMS Bacapedia<sup>+</sup></h4>
35           )}
36         <button
37           className="toggle-btn"
38           onClick={toggleSidebar}
39           aria-label="Toggle Sidebar"
40         >
41           <i
42             className={`bi ${isCollapsed ? "bi-chevron-right" : "bi-chevron-left"}`}
43           ></i>
44         </button>
45       </div>
46       <div className="sidebar-content">
47         <ul className="sidebar-menu">
48           {Menu_CMS.map((item) => (
49             <li key={item.path}>
50               <Link
51                 href={item.path}
52                 className={pathname === item.path ? "active" : ""}
53                 data-title={item.name}
54               >
55                 <i className={item.icon}></i>
56                 <span className="menu-text">{item.name}</span>
57               </Link>
58             </li>
59           ))}
60         </ul>
61       </div>
62     </aside>
63   );
64 }
```

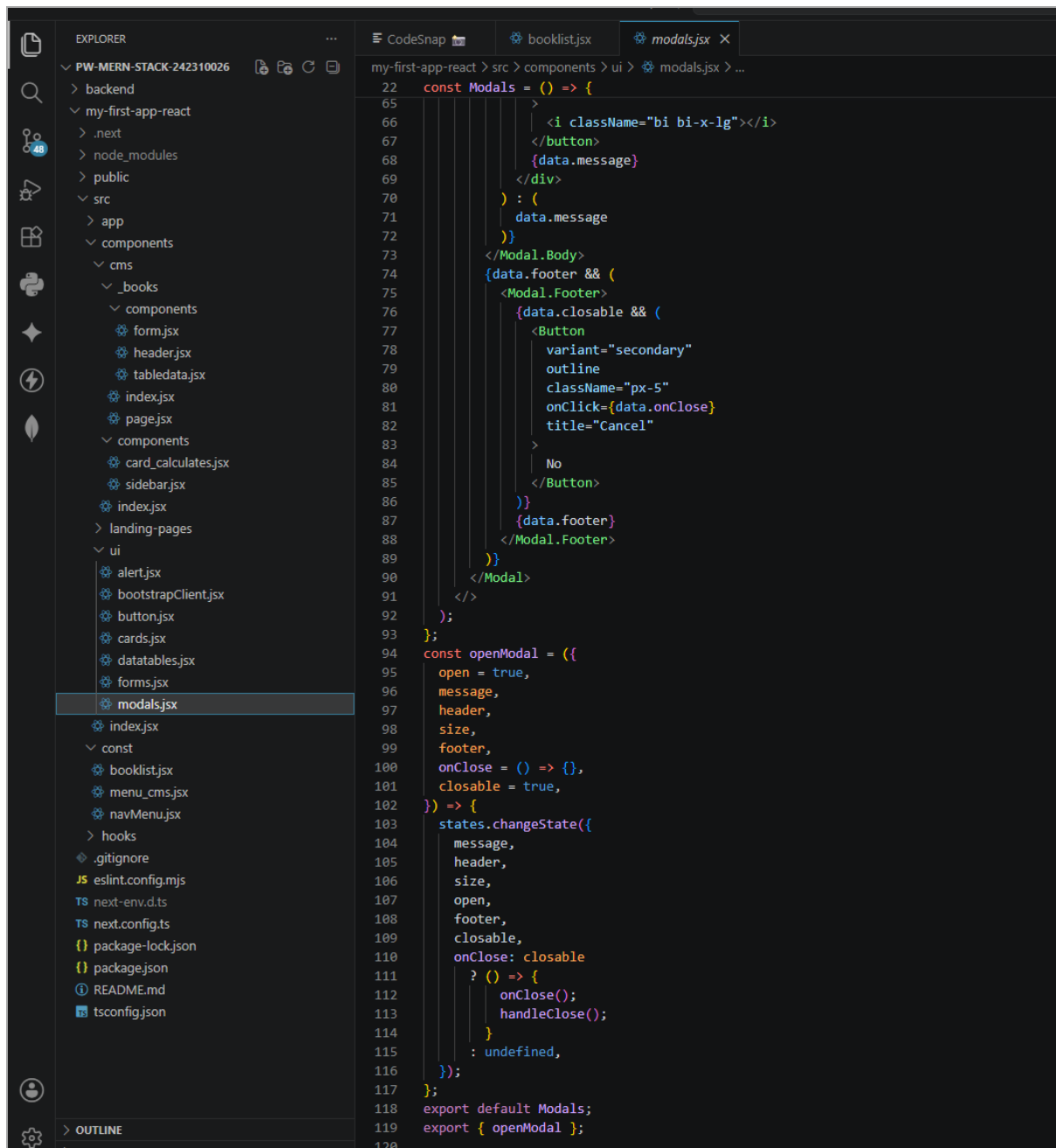
src/components/ui/ (buttons, alert, forms, cards, modals, datatables)

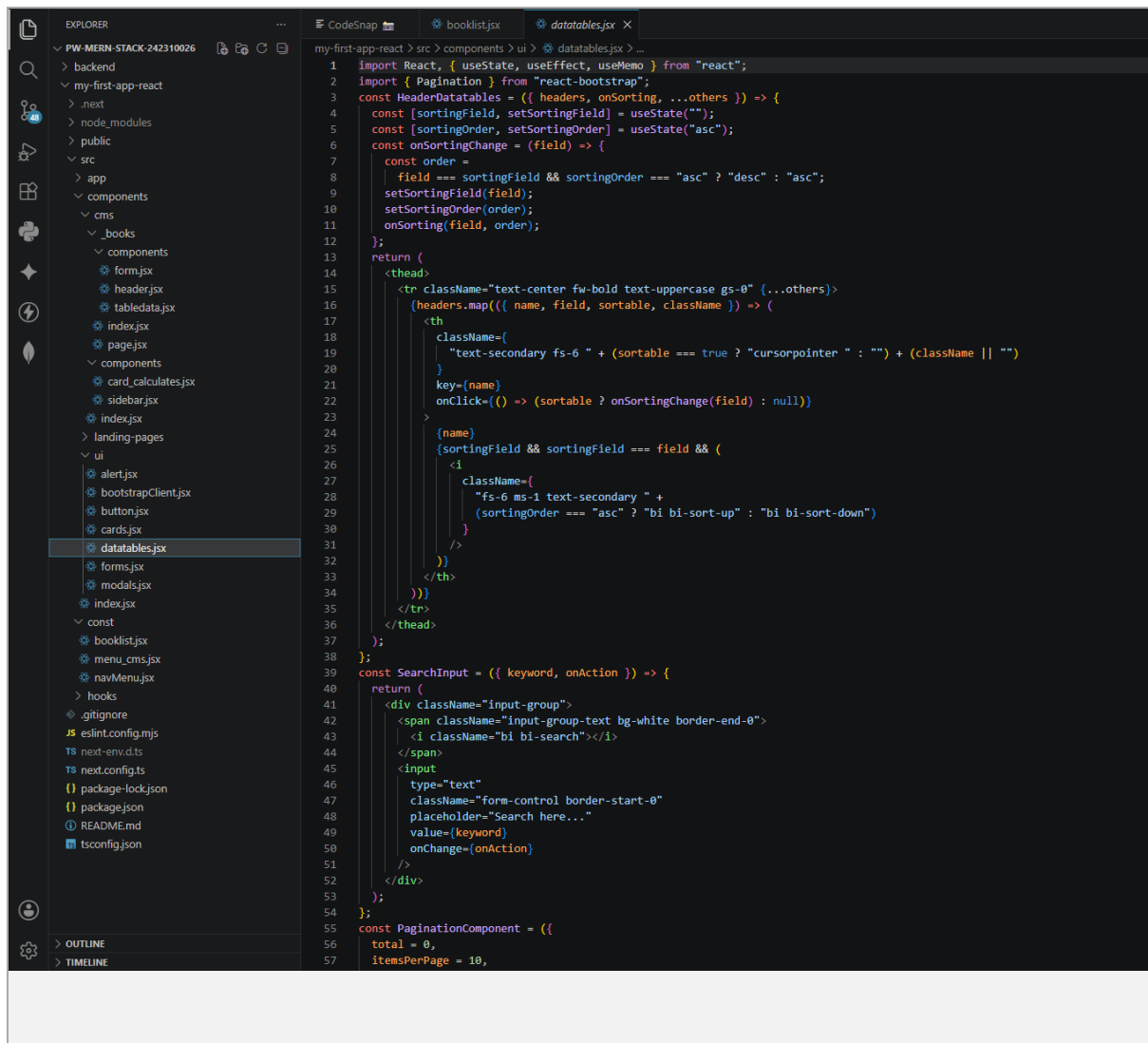


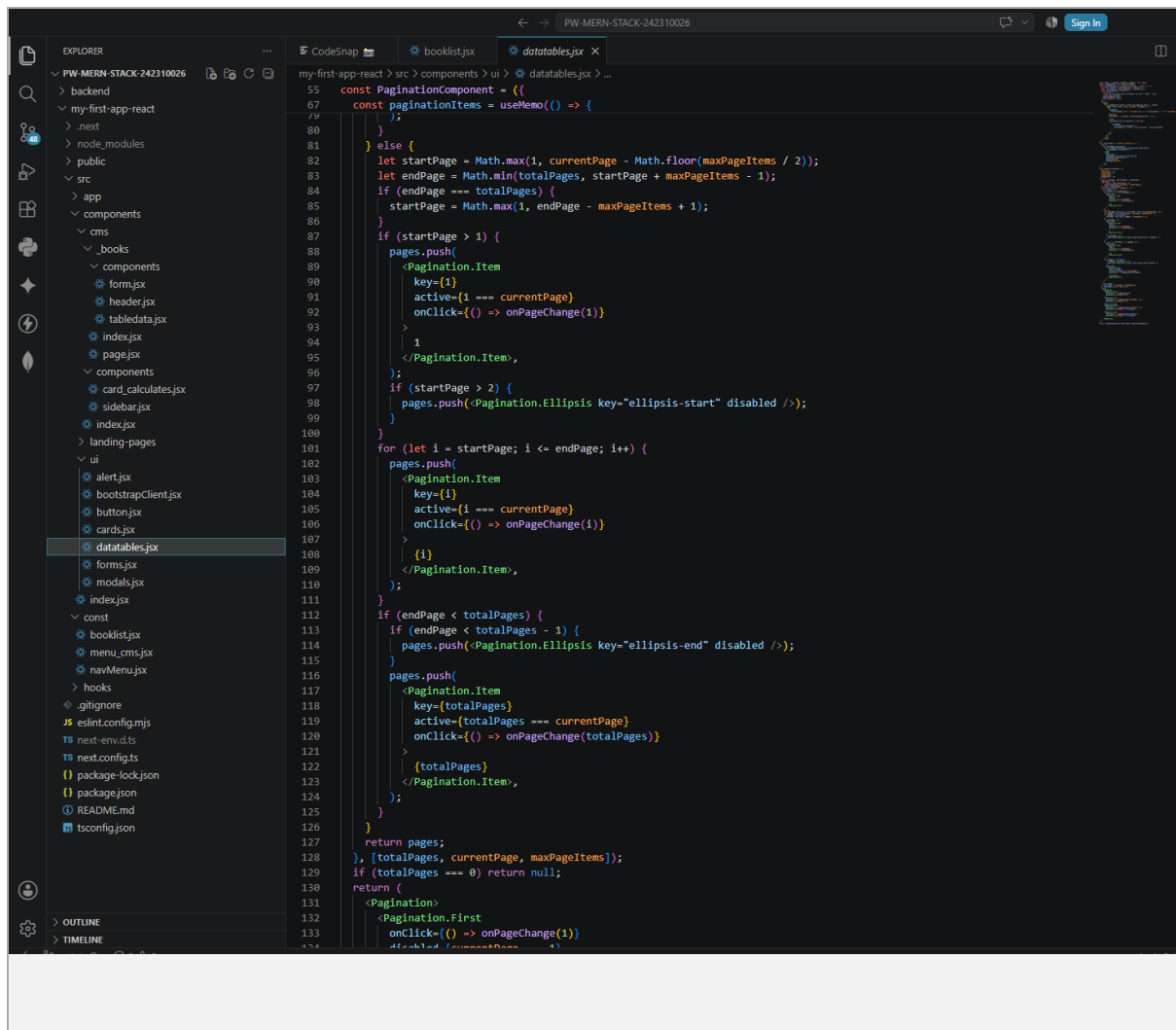








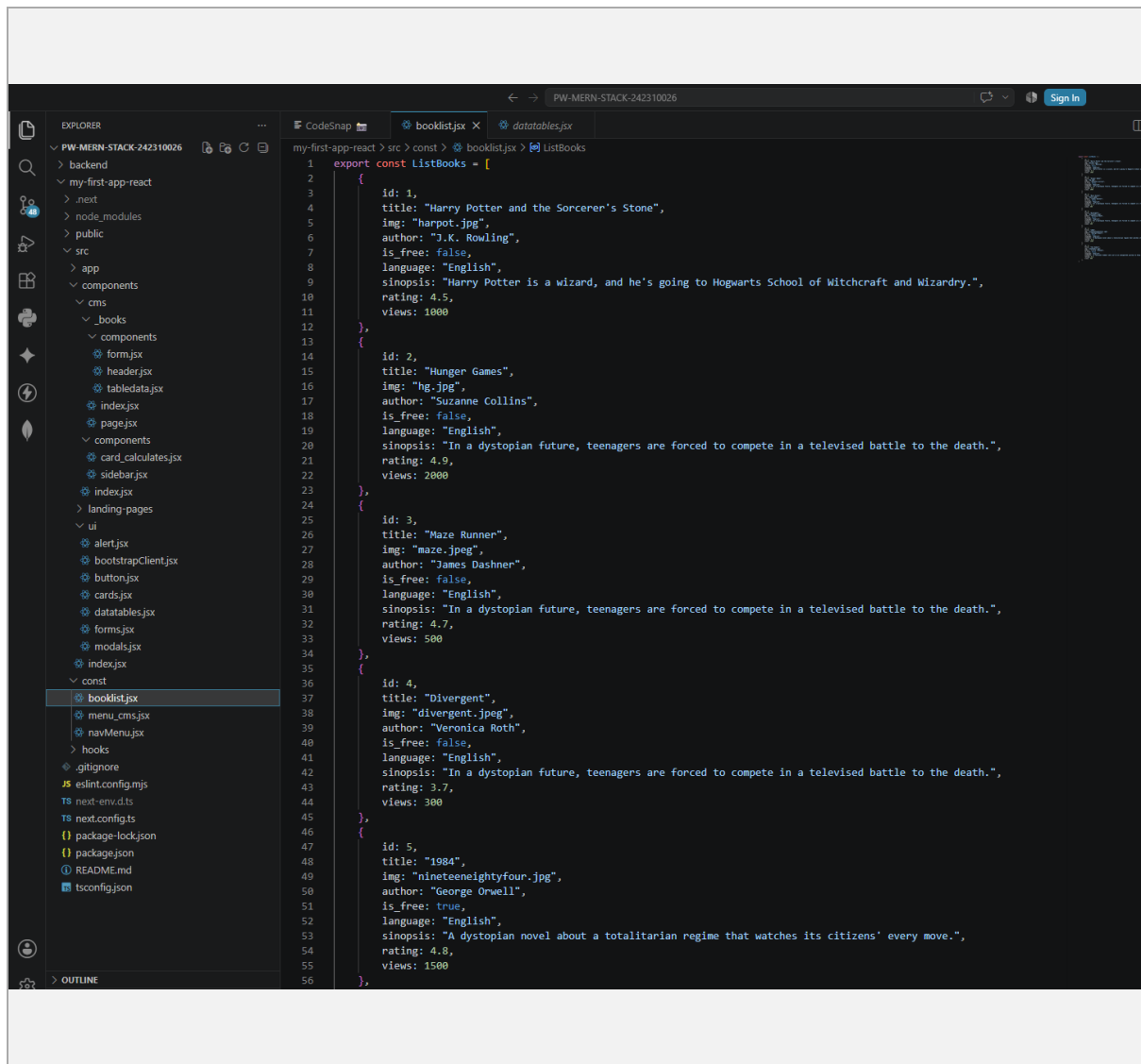


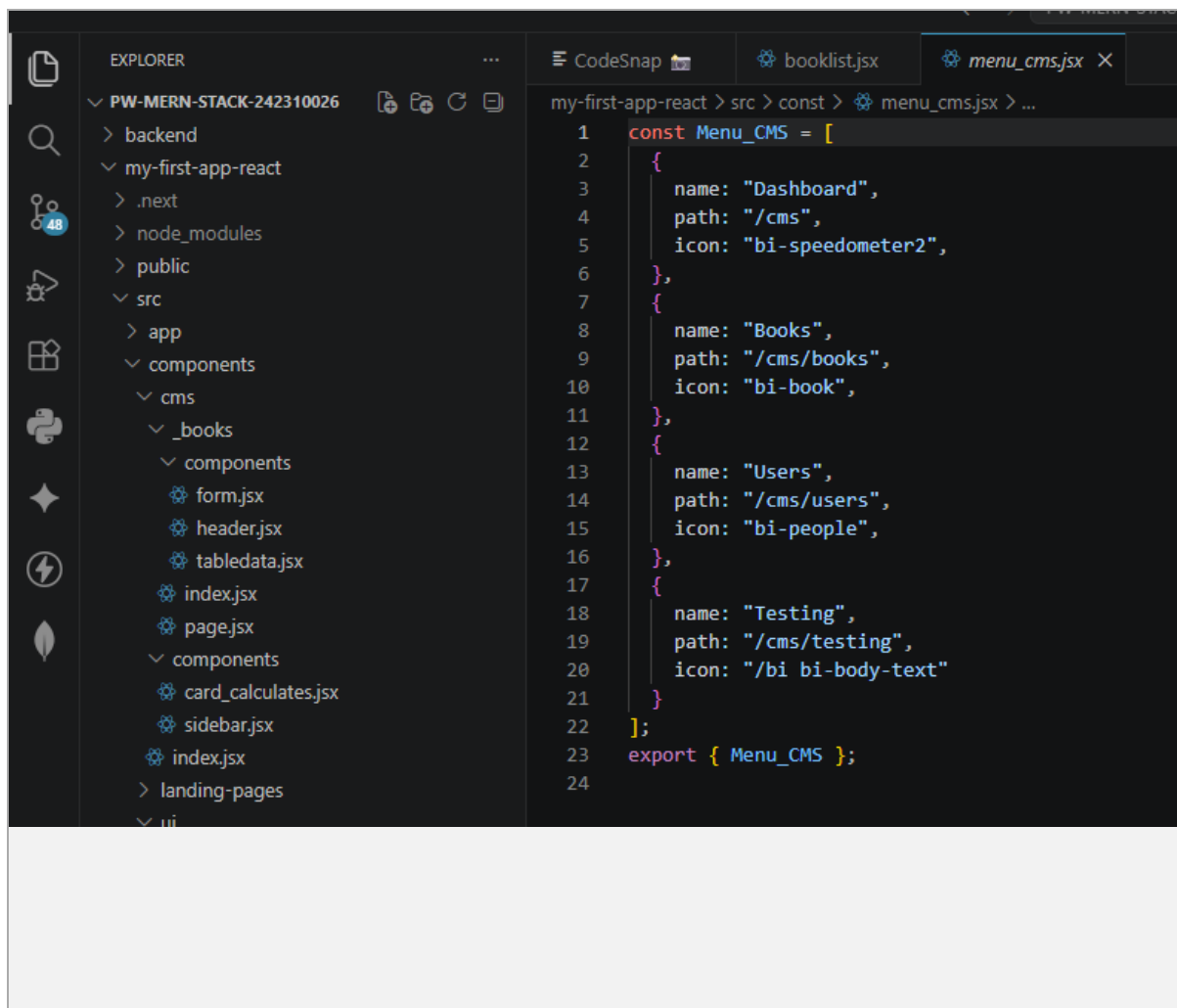


The screenshot shows a VS Code editor with a project named 'PW-MERN-STACK-242310026'. The Explorer sidebar on the left shows the project structure, with 'src/components/ui/datatables.jsx' selected. The main editor displays the code for '/datatables.jsx', which defines a 'PaginationComponent' using 'useMemo' and 'useState' hooks. The component renders a pagination interface with 'Pagination.Item', 'Pagination.First', 'Pagination.Prev', 'Pagination.Next', and 'Pagination.Last' components. The code includes logic for calculating the total number of pages, the current page, and the range of pages to display. The 'Pagination.Item' component is used to render individual page numbers, and the 'Pagination.First', 'Pagination.Prev', 'Pagination.Next', and 'Pagination.Last' components are used to render navigation links. The code is as follows:

```
55 const PaginationComponent = ({
67   const paginationItems = useMemo(() => {
100   }
101   for (let i = startPage; i <= endPage; i++) {
102     pages.push(
103       <Pagination.Item
104         key={i}
105         active={i === currentPage}
106         onClick={() => onPageChange(i)}
107       >
108         {i}
109       </Pagination.Item>,
110     );
111   }
112   if (endPage < totalPages) {
113     if (endPage < totalPages - 1) {
114       pages.push(<Pagination.Ellipsis key="ellipsis-end" disabled />);
115     }
116     pages.push(
117       <Pagination.Item
118         key={totalPages}
119         active={totalPages === currentPage}
120         onClick={() => onPageChange(totalPages)}
121       >
122         {totalPages}
123       </Pagination.Item>,
124     );
125   }
126   }
127   return pages;
128 }, [totalPages, currentPage, maxPageItems]);
129 if (totalPages === 0) return null;
130 return (
131   <Pagination>
132     <Pagination.First
133       onClick={() => onPageChange(1)}
134       disabled={currentPage === 1}
135     />
136     <Pagination.Prev
137       onClick={() => onPageChange(currentPage - 1)}
138       disabled={currentPage === 1}
139     />
140     {paginationItems}
141     <Pagination.Next
142       onClick={() => onPageChange(currentPage + 1)}
143       disabled={currentPage === totalPages}
144     />
145     <Pagination.Last
146       onClick={() => onPageChange(totalPages)}
147       disabled={currentPage === totalPages}
148     />
149   </Pagination>
150 );
151 };
```

src/const/ (bookList.jsx, menu_cms.jsx)





src/hooks/useScrolled.jsx

